Emission and Readout — Phase, Momentum, Gauge and Metastability

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How a settled field becomes an ordered output, and the mechanisms that stop it locking.

Read This Before Citing

The emission problem is only half solved, and the two halves are easy to confuse.

MRS (`mrs.md`) has a *recognition* half and an *emission* half. Recognition works: bigram sensory lanes survive misspellings and transpositions at 7/7 measured. Emission **babbles** and is retired. Citing MRS as a working generator misreports it.

Concept-level ordering works; byte-level does not. CPME's phase-collapse sweep and CPSB's gauge offsets order *which concept follows which*, and that holds. Producing the *bytes of a word* is unsolved, and the Language volume argues that for grammar the question was

mis-stated rather than merely hard.

CSAT is opt-in and off by default.

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Escaping the Echo: Momentum-Coupled Phase Emission for Generative Resonance in Synthetic Intelligence

Abstract In wave-based cognitive substrates utilizing Continuous Morphological Dynamics (CMD), external stimuli inherently generate maximal amplitude at the point of injection, leading to "Stimulus Reflection" (the parrot effect) during syntax emission. We propose a novel non-algorithmic architecture, **Momentum-Coupled Phase Emission (MCPE)**, which transitions the readout mechanism from absolute amplitude to kinetic wave momentum. By coupling syntax oscillators exclusively to the first derivative of the energy field ($dA/dt > 0$), the substrate naturally ignores decaying direct stimuli and phase-locks exclusively to emergent, forward-scattering internal thoughts.

1. Introduction: The Acoustic Blinding Problem

In the current iteration of the SI engine, the Kuramoto syntax slots (Subject, Verb, Object) phase-lock to cavities with the highest absolute amplitude (A). When external input (e.g., "notch created minecraft") drives the field, those specific cavities reach maximum A. Before the wave can propagate deep into the latent graph to retrieve a meaningful association, the syntax slots lock onto the massive input wave, resulting in a mirrored output. The system is structurally blind to its own internal resonance because the external drive acts as an acoustic flashbang.

2. Novel Architecture: Momentum-Coupled Phase Emission (MCPE)

To solve this without algorithmic filtering, we introduce a Hamiltonian shift in the readout layer. Instead of coupling syntax slots to the "position" (Amplitude, A) of the wave, we couple them to the "momentum" (Flux, P).

2.1. First-Derivative Readout (dA/dt Gating) In a physical wave system, a cavity directly hit by a stimulus reaches peak energy and immediately begins to disperse it (cooling down). Its amplitude is high, but its momentum is negative ($dA/dt \leq 0$). Conversely, deep-graph nodes that are currently being activated by internal resonance are absorbing energy; their amplitude might be lower, but their momentum is strictly positive ($dA/dt > 0$).

We redefine the coupling strength K_{emit} between a cavity i and a syntax slot as:

$$K_{emit} = \Theta\left(\frac{dA_i}{dt}\right) \cdot A_i$$

Where $\Theta(x)$ is the Heaviside step function (returns x if $x > 0$, else 0). *Result:* Directly driven words (the user's prompt) instantly decouple from the output slots the millisecond the prompt ends, because they are decaying. The output slots are physically forced to lock onto the *ripples* (the internal thoughts), not the *splash* (the user's input).

2.2. Thermodynamic Refractoriness (The "Acoustic Mass") To prevent the wave from bouncing directly back to the input nodes, we modify the cavity inertia. When a cavity is driven externally, it accumulates temporary "acoustic mass" M_i:

$$\frac{dM_i}{dt} = \gamma I_{ext} - \delta M_i$$

Higher mass lowers the cavity's fundamental oscillating frequency temporarily, pushing it out of the L6-A (3Hz) emission band. It retains its associative links but becomes too "heavy" to be picked up by the Kuramoto slots.

2.3. Temporal Settle Separation (Perception vs. Articulation) We separate the physics tick into two distinct phases using a global damping coefficient ζ:

1. **Injection Phase (T_0 to T_1):** External wave hits. ζ is high. Energy is absorbed. Syntax slots are decoupled ($K = 0$).
2. **Articulation Phase (T_1 to T_2):** External drive is 0. ζ drops. The wave propagates through the L6-A bridges. Syntax slots engage, seeking positive momentum ($dA/dt > 0$).

3. Expected Dynamics

Scenario: User inputs `insulin controls glucose`.

1. T_0 **(Injection):** Cavities `insulin`, `controls`, and `glucose` spike in absolute amplitude ($A \approx 1.0$). Their mass M_i increases.
2. T_1 **(Propagation):** Input ceases. The energy from `insulin` and `glucose` propagates through the ontology graph to connected cavities like `metabolism`, `pancreas`, and `glycogen`.
3. T_2 **(Articulation):**

- `insulin` has high A, but $dA/dt < 0$ (cooling). It is ignored.
- `pancreas` has medium A, but $dA/dt > 0$ (warming up). It is locked.
- The Kuramoto slots phase-lock to `pancreas regulates metabolism` rather than parroting the input.

Conclusion: MCPE preserves the zero-rule continuous dynamics of the substrate while physically guaranteeing that the engine generates consequences rather than echoing stimuli.

Holographic Topological Resonance (HTR): Shifting from Predictive Generation to Structural Comprehension in Wave-Based Synthetic Intelligence

Abstract Current Large Language Models (LLMs) simulate comprehension through autoregressive statistical approximation, compressing language into latent vectors without spatial or structural understanding. We propose **Holographic Topological Resonance (HTR)**, a wave-based cognitive architecture for true language comprehension and construction. By treating sentences not as token sequences but as **chiral wave-interference patterns**, the engine physically "understands" by aligning input topology with intrinsic graph geometry. Language construction is achieved via **Phase-Collapse Projection**, unfolding an n-dimensional thought back into a 1D temporal sequence without templates, syntax trees, or stochastic prediction.

1. Introduction: The Illusion of Prediction

An LLM predicting "Minecraft" after "Notch created" is analogous to a parrot mimicking a frequent sound sequence. It lacks the internal geometry of *creation* or the structural boundary between *agent* and *artifact*. True understanding requires an internal isomorphic state—a physical subgraph where "Notch" and "Minecraft" are held in a dynamic, tensioned relationship by a directed force ("created"). To bridge the gap from a 1D string of user bytes to an n-dimensional cognitive state, and back to a 1D response, we introduce the HTR architecture.

2. Novel Architecture: The HTR Substrate

To achieve true comprehension, the engine must perform three non-algorithmic physical processes:

2.1. Chiral Wave Injection (Encoding Syntax as Phase-Delay) A sentence has directionality (e.g., "Dog bites man" vs. "Man bites dog"). Instead of parsing arrays or using positional embeddings, HTR uses **Chiral (Directional) Wave Injection**. As the user's bytes stream into the engine, they hit the substrate with sequential time delays (Δt). This creates a directional wave-front.

- **Mechanism:** The time delay translates directly into a **Phase Angle (ϕ)**.

$$\phi_i = \omega_{base} \cdot (T_{current} - t_{arrival_i})$$

- **Chirality convention (explicit).** With this definition, $(T_{current} - t_{arrival_i})$ is the elapsed time since byte i entered the substrate. Earlier arrivals therefore accumulate more phase; later arrivals accumulate less. "Subject leads in phase" means the subject cavity is *ahead* in the phase progression because it entered the substrate first, not that it has the numerically smaller phase value. Implementations must record $t_{arrival_i}$ per cavity at drive time and evaluate ϕ_i from the running $T_{current}$; any sign inversion of the elapsed-time term will invert the encoded chirality and produce reversed word order in the readout stage.
- **Result:** The input is no longer a list of words. It is a standing interference pattern where the *Subject* leads in phase, the *Verb* is phase-shifted by $\pi/2$, and the *Object* by π. Syntax is natively encoded as a physical gradient across the graph.

2.2. Topological Isomorphism (Structural Resonance Metric) A prompt is said to *resonate structurally* with the engine's long-term memory when the temporary interference pattern of the prompt aligns with the physical shape of a stored axiom subgraph. This is a strictly weaker claim than "the engine understands the prompt" — resonance is a necessary, not sufficient, correlate of comprehension — but it is directly measurable in the substrate's existing state and requires no additional variables.

- **Mechanism:** The injected chiral wave propagates through the graph. The L6-A bridges act as resonance chambers. If the user asks a complex question (e.g., "What lowers blood sugar?"), the wave excites "lowers" and "blood sugar".
- Because of the Hebbian physics, the wave naturally converges on "insulin" or "metformin" not because they are statistically next, but because they physically complete the topological circuit.
- **Resonance metric.** Let M_{touched} be the set of macro-nodes whose constituent cavities have been reached by the current input wavefront (i.e., whose `quiet_ticks` is below the idle threshold). Define the *localized* Kuramoto order parameter:

$$R_{\text{local}} = \frac{\left|\sum_{m \in M_{\text{touched}}} A_m \, e^{i\theta_m}\right|}{\sum_{m \in M_{\text{touched}}} A_m + \varepsilon}.$$

  The comprehension-correlated signal is the *rise* in this quantity above its recent EMA baseline:

$$\Delta R = R_{\text{local}} - \text{EMA}(R_{\text{local}}).$$

  A significant $\Delta R > 0$ after propagation indicates the input drove the field into a phase-consensus state substantially more ordered than baseline drift. Absolute R is not diagnostic: a trivially quiescent field can have $R \approx 1$ with no semantic content. Only the localized rise ΔR, restricted to macros the input actually reached, is meaningful.
- **Interpretation.** ΔR is best read as a *structural resonance metric*, not a proof of understanding. High ΔR licenses articulation; low ΔR licenses honest silence. Anything stronger requires a phase-aware Hebbian consolidation rule (see §5, Failure Modes), which is proposed as future work.

2.3. Phase-Collapse Projection (Emergent Language Construction) Once the internal subgraph reaches a stable topological state (the "thought"), it must be translated back into language. Unlike LLMs that predict word-by-word, HTR *projects* the entire structure into a 1D temporal sequence by unwinding the phase space.

- **Per-cavity phase.** Rather than inheriting phase from the macro-node cluster (which would collapse an entire cluster's cavities to the same emission instant), each cavity is assigned an *instantaneous phase* derived from its own two-dimensional state:

$$\phi_i = \text{atan2}(v_i, \; \omega_{\text{base}} \cdot x_i).$$

  This uses no additional state — velocity v_i and displacement x_i are already integrated by the substrate — and yields a continuous phase field over the cavity population with fine enough resolution to unwind a several-node subgraph into distinct emission instants.
- **Mechanism.** A carrier phase ϕ_{carrier} is swept monotonically from 0 to 2π (non-cyclic, one pass, wrap-around terminates the sweep). At each swept instant, each cavity's emission weight is a Gaussian window on the phase difference:

$$w_i(\phi_{\text{carrier}}) \;=\; A_i \cdot \exp\!\left(-\kappa_{\text{sweep}} \cdot (\phi_{\text{carrier}} - \phi_i)^2\right).$$

  The Dirac δ-form is a limiting idealisation; the implementation uses the Gaussian directly to avoid the discontinuity artefact and to give a physically-motivated window width $1/\sqrt{2\kappa_{\text{sweep}}}$.
- **Emission trigger.** At each sweep step the cavity with the highest w_i that also satisfies the momentum eligibility (`ReLU(dA_i/dt) > 0`, from the MCPE readout layer) and the emission floor is emitted, then suppressed from all subsequent steps in the same sweep.

- **Result.** The thought is an n-dimensional physical structure. The carrier phase acts as a 1D scanner sweeping across it. The nodes emit in order determined by their per-cavity phases; the output length is emergent — as many tokens as the field has distinct high-momentum peaks under the sweep — and can be shorter than the previous fixed slot count when the field is sparse, or longer when it is dense. No syntax trees, no `if-else` grammar rules, no next-token probabilities. The sentence constructs itself through geometric projection.

3. Expected Dynamics: Processing a Complex Query

User Prompt: `metformin activates what in cells`

1. **Injection (Chirality):** `metformin` hits first ($\phi \approx 0$), `activates` second ($\phi \approx \pi/3$), `cells` third ($\phi \approx 2\pi/3$). A directional energy gradient flows from metformin outward.
2. **Comprehension (Topology):** The gradient flows through the `REL_ACTIVATES` tunnels. It hits `AMPK`. Because `AMPK` is structurally linked to `cells` in the long-term graph, a resonance loop completes. The substrate physically stabilizes. **The engine has understood the query.**
3. **Construction (Collapse):** The carrier wave sweeps from 0 to 2π.

- At ϕ_{early}, it hits the resonating result (`ampk`).
- At ϕ_{mid}, it hits the relational tunnel (`regulates`).
- At ϕ_{late}, it hits the downstream effect (`metabolism`). **Output Generated:** `ampk regulates metabolism`.

Conclusion: The HTR architecture entirely bypasses statistical generation. By encoding word order as arrival-phase chirality, structural match as a localized rise in Kuramoto coherence, and generation as continuous phase-collapse projection over a per-cavity phase field, the substrate realizes input encoding, internal state metric, and output construction on a single phase manifold. Whether this constitutes *comprehension* in a philosophically loaded sense is deferred to future work; what the mechanism does provide is a substrate whose emission cannot be reduced to next-token statistical approximation because the emission is *geometric*, not *distributional*.

---

5. Failure Modes

Three failure modes are known and should be documented explicitly before any implementation is deployed.

1. **Phase clumping without per-cavity resolution.** Prior CMD implementations gave cavities the phase of their parent macro-cluster. Under such a scheme Phase-Collapse Projection degenerates to macro-level emission: every cavity in a cluster emits at the same sweep instant, and the output collapses to at most one token per macro. The per-cavity phase definition $\phi_i = \text{atan2}(v_i, \omega_{\text{base}} \cdot x_i)$ from §2.3 is a required precondition; implementations that skip it will observe uniform bursts rather than sequential emission.

2. **Chirality–Hebbian mismatch.** Chiral phase encoding on injection (§2.1) only contributes to comprehension if the long-term memory is *also* consolidated with phase awareness. The Hebbian rule as inherited from the CMD substrate updates tunnel plasticity from the product of endpoint velocities and knows nothing about phase alignment. A phase-coincidence Hebbian term of the form

$$\Delta w_{ij} \propto v_i \cdot v_j \cdot \cos(\theta_i - \theta_j)$$

is the natural companion, and is proposed as a separate follow-up ("HTR-II: Phase-Consistent Consolidation"). Without it, the chirality of injection dissipates over turns rather than being stored, limiting the resonance metric's sensitivity to recent context.

3. **Circular sweep ambiguity and empty resonance.** The Gaussian sweep window is unimodal on the real line, but phase is defined modulo 2π: a cavity with ϕ_i near 0 or near 2π can pattern-match at either boundary of the sweep, producing duplicate or edge-clustered emissions. The non-cyclic sweep $[0, 2\pi)$ prescribed in §2.3 mitigates this by treating wrap-around as end-of-output. Empty resonance — the case where the graph is too sparse or the input too disconnected for ΔR to rise — produces zero emissions after the momentum and floor filters; this is the correct honest-silence behaviour but will initially read as a regression to users habituated to the fixed-slot readout's always-something-out property.

6. Relation to Momentum-Coupled Phase Emission (MCPE)

HTR and the concurrently-proposed MCPE architecture are complementary rather than competing. MCPE identifies *which* cavities are eligible to emit (positive-momentum, low acoustic-mass, active only during the articulation phase of the settle split). HTR identifies *in what order* eligible cavities should emit (per-cavity phase, swept by a carrier). The two mechanisms operate on orthogonal observables: MCPE on dA_i/dt, M_i, and drive envelope; HTR on ϕ_i, R_{local}, and the arrival-phase field. A substrate implementing both simultaneously — see the joint CPME proposal — is expected to exhibit both non-parroting behaviour (from MCPE's momentum filter) and emergent variable-length output in structurally-resonant word order (from HTR's phase-collapse projection).

Chiral Phase-Momentum Emission (CPME)

A unified emission substrate combining the Momentum-Coupled Phase Emission (MCPE) filter architecture with the Holographic Topological Resonance (HTR) phase-projection readout. CPME operates on a single phase manifold: chirality on injection, coherence-rise as the resonance metric, momentum as the eligibility filter, and continuous phase-collapse sweep as the emission mechanism. No syntax trees, no grammar rules, no next-token statistics, and no fixed output length.

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0. Position

The Continuous Morphological Dynamics (CMD) substrate, extended by Voltage-Gated Harmonic Topology (VGHT), produces domain-orthogonal cognition with sparse cross-domain analogical bridging. Its emission layer, however, has two orthogonal deficits that the previous literature treats separately:

1. **Stimulus Reflection.** External input directly drives concept cavities to maximal amplitude; the fixed-slot Kuramoto readout locks onto those cavities before internal propagation can produce a downstream response. The system parrots.
2. **Fixed-Slot Rigidity.** The Subject/Verb/Object slot readout presupposes a three-word template. Simpler answers waste slots; more complex answers are truncated. Word order is imposed by the slot definitions rather than emerging from the field.

MCPE addresses (1) by shifting the readout observable from amplitude A to momentum dA/dt and by muting recently-driven cavities via acoustic mass. HTR addresses (2) by replacing the fixed slots with a continuous carrier-phase sweep over a per-cavity phase field, and by introducing arrival-time chirality on injection so that word order becomes a physical gradient rather than a syntax rule.

CPME unifies the two proposals into a single emission substrate. MCPE identifies *which* cavities are eligible to emit; HTR identifies *in what order* eligible cavities emit. The two operate on orthogonal state variables and compose without collision.

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1. Architecture

Let the substrate be the CMD + VGHT engine as previously specified: cavities indexed by $i \in [0, N)$ with per-cavity displacement x_i, velocity v_i, heat T_i, and quiet age q_i; tunnels with per-tunnel stiffness k, damping c, polarity p, and voltage gating $g \in \{0, 1\}$; and macro-nodes with per-cluster phase θ_m, amplitude A_m, and natural frequency ω_m drawn from the VGHT sub-band ladder. The base cavity frequency is ω_0.

CPME adds six mechanisms.

1.1 Chiral Wave Injection (from HTR §2.1) Each cavity records the simulation time at which it was most recently driven:

```
t_arrival_i ← sim_time    (whenever drive(i, ·) is called)
```

The cavity's oscillator phase at any subsequent tick is $\phi_i^{\text{driven}} = \omega_0 \cdot (T_{\text{current}} - t_{\text{arrival},i})$. Earlier-arriving bytes therefore accumulate *more* phase; the subject leads in phase because it entered the substrate first. Chirality is inherited *for free* from the existing sinusoidal drive term — the arrival time only needs to be *recorded* so the readout can reference it. No new tick math.

1.2 Momentum Filter (from MCPE §2.1) Every cavity carries an exponential-moving-average amplitude envelope:

`E_i` $\leftarrow \beta \cdot$ `E_i + (1 −` β`) · |x_i|` (β `= 0.9`)

and the tick publishes the *previous-tick* envelope as E_i^{prev} so that the momentum

`P_i = E_i − E_i^{\text{prev}}`

is the low-pass-filtered rate of change of amplitude. Directly-driven cavities that have peaked and are dispersing energy have $P_i \leq 0$; deep-graph cavities absorbing energy from downstream propagation have $P_i > 0$. The emission eligibility factor is $\text{ReLU}(P_i)$, a smooth non-negative scalar.

This replaces the naïve $dA/dt = |v|$ - style derivative, which oscillates at $2\omega_{\text{drive}}$ and produces a strobe artefact in the readout. Envelope EMA is the physically correct low-pass.

1.3 Acoustic Mass (from MCPE §2.2) Each cavity carries a per-cavity mass increment M_i driven by the external injection load:

`dM_i/dt =` $\gamma \cdot$ `I_{ext, i} −` $\delta \cdot$ `M_i`

with $\gamma = 0.30$, $\delta = 0.08$. $I_{\text{ext},i}$ is the instantaneous drive envelope on cavity i, i.e., precisely the signal that makes an input-driven cavity "loud." The resulting emission mask

`μ_i = 1 / (1 + M_i / M_{scale})` (`M_scale = 1.0`)

falls toward zero for freshly-driven cavities and returns toward $\mu_i = 1$ as M_i decays. Applied only in the emission scoring; the field dynamics are untouched, so graph connectivity is preserved.

1.4 Settle Separation (from MCPE §2.3) The tick maintains a global injection load

`L =` Σ`_i drive_amp_i · drive_envelope_i`

and derives a *slot coupling* factor

`K_settle = 1 −` σ`((L − L_thresh) / L_scale)`

with σ the logistic, $L_{\text{thresh}} = 0.5$, $L_{\text{scale}} = 0.2$. During active injection, $K_{\text{settle}} \approx 0$ and the emission readout is suppressed. As the drive envelopes decay, $K_{\text{settle}} \to 1$ and articulation engages. The transition is continuous — no discrete $T_0/T_1/T_2$ boundary.

1.5 Coherence-Rise Metric (from HTR §2.2) The tick computes the *localized* Kuramoto order parameter over input-touched macros:

`R_local = |`Σ`_{m ∈ touched} A_m ·` $e^{i\theta_m}$`| / (`Σ`_{m ∈ touched} A_m +` ε`)`

where M_{touched} is the set of macros with at least one non-quiet cavity ($q_i < K_{\text{idle}}$). R_{local} is tracked against its EMA baseline $\bar{R}$:

Δ`R = R_local − R̄` (`R̄ ←` λ`R̄ + (1−`λ`) R_local,` λ `= 0.98`)

ΔR is exposed as a substrate observable but does not directly gate emission; it is diagnostic. Silence when ΔR is low emerges from the emission floor being unmet, not from an explicit ΔR gate. This preserves the substrate's honest-silence property without turning ΔR into a rule.

1.6 Phase-Collapse Projection (from HTR §2.3) The emission readout is replaced entirely. Per-cavity phase is computed on demand from existing state:

φ`_i = atan2(v_i,` $\omega_0 \cdot$ `x_i)`

The carrier phase ϕ_{carrier} is swept monotonically across $[0, 2\pi)$ in K_{steps} discrete increments (default $K_{\text{steps}} = 12$). At each swept instant, the cavity chosen for emission is the arg-max of

```
w_i(φ_carrier) = A_i · ReLU(P_i) · μ_i · K_settle
               · exp(−κ_sweep · (φ_carrier − φ_i)^2)
```

where $A_i = |x_i|$ is the instantaneous amplitude and $\kappa_{\text{sweep}} = 4$ sets the Gaussian window width. Winners below the floor $\theta_{\text{emit}} \cdot \max_j w_j$ are silenced (no emission at that sweep step). Winners are suppressed from all subsequent steps of the sweep (each cavity emits at most once per turn). The sweep terminates non-cyclically at $\phi_{\text{carrier}} = 2\pi$, so wrap-around is not a source of duplicate emissions.

Output length is emergent: as many tokens as there are distinct high-weight peaks under the sweep. A sparse field emits few tokens (possibly zero — honest silence); a dense, structurally-resonant field emits many.

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2. Composition

The six CPME mechanisms and the prior CMD + VGHT stack all operate on distinct observables and compose additively:

Layer	Mechanism	State variable
Substrate	CMD Curie gate	T_i
Substrate	VGHT sub-bands	ω_m
Substrate	VGHT voltage gate	$\|x_a\| \cdot \|x_b\|$
Substrate	VGHT viscosity	E_{total}
Injection	**CPME chirality**	$t_{\text{arrival},i}$
Filter	**CPME momentum**	E_i, E_i^{prev}
Filter	**CPME acoustic mass**	M_i
Filter	**CPME settle**	L
Metric	**CPME coherence rise**	$R_{\text{local}}, \bar{R}$
Emission	**CPME phase collapse**	$\phi_i, \phi_{\text{carrier}}$

The tick loop retains its no-if/else contract: chirality is a recorded scalar, momentum is EMA arithmetic, acoustic mass is Ornstein–Uhlenbeck, settle coupling is a sigmoid, coherence is a complex-sum norm, and phase collapse is a Gaussian window over a swept center. All are pure scalar or complex-scalar operations.

---

3. Emission Contract

At the end of each turn (i.e., after propagation ticks have run), the substrate exposes a single function:

`emit()` → `ordered list of cavity ids`

whose implementation is the Phase-Collapse sweep of §1.6. There is no `readout_slot_winners()` remaining, no S/V/O slot layer, no `syntax_slots_enabled` flag. Word order is arg-max under the Gaussian sweep window; slot count is not a parameter.

Backwards compatibility is preserved as a thin adapter: callers that previously expected a three-element slot list receive the first three emissions of the sweep, padded with `-1` if fewer are produced.

---

4. Parameters

Symbol	Default	Meaning
β	0.9	Momentum EMA smoothing factor
γ	0.30	Acoustic mass accumulation rate
δ	0.08	Acoustic mass decay rate
M_{scale}	1.0	Acoustic mask half-point
L_{thresh}	0.5	Settle-separation threshold
L_{scale}	0.2	Settle-separation sharpness
λ	0.98	Coherence EMA baseline decay
K_{steps}	12	Phase-collapse sweep resolution
κ_{sweep}	4.0	Gaussian window sharpness
θ_{emit}	0.05	Emission floor as fraction of peak w

Ten parameters, all continuous, all with default values that recover CMD-only behaviour when set to their extreme values (e.g., $K_{\text{steps}} = 3$, $\gamma = 0$, $\kappa_{\text{sweep}} \to \infty$ reduce CPME to the three-slot readout).

---

5. Expected Dynamics

Prompt. `metformin activates receptor`.

1. **Injection** ($t = 0$ through $t_{\text{arrival, receptor}} + \Delta$): `metformin`, `activates`, `receptor` cavities are driven in that temporal order. Their t_{arrival} values differ by the inter-byte interval, encoding the chirality gradient. Acoustic mass M grows on each; their emission masks μ approach zero. Injection load L is high, so $K_{\text{settle}} \approx 0$ — the emission readout is silent regardless of field state.
2. **Propagation** (~10-40 ticks): Drive envelopes decay. Energy propagates through the graph's `REL_ACTIVATES` tunnels and VGHT voltage-gated bridges. Downstream cavities (`ampk`, `regulates`, `metabolism`, `pancreas`, etc., depending on ontology) rise in amplitude. Because they are absorbing energy from propagation rather than external drive, their momentum $P > 0$ and their acoustic mass $M \approx 0$.
3. **Settle** (~40-80 ticks): Drive envelopes decay to near zero. L falls below threshold, $K_{\text{settle}} \to 1$. Localized coherence R_{local} has risen well above $\bar{R}$; $\Delta R > 0$ signals structural resonance.
4. **Emission** (phase-collapse sweep): The carrier phase sweeps $[0, 2\pi)$. Directly-driven cavities have $\mu \approx 0$ and are excluded from arg-max even if their phase would align. Downstream cavities with $P > 0$ and $\mu \approx 1$ compete on $\phi_i \approx \phi_{\text{carrier}}$; the ordering of their phases (from their integrated positions and velocities) becomes the emission order. Output: three to four tokens describing the *consequence* of the query, not the query itself.

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6. Failure Modes

Five known modes, all inherited from the parent proposals:

1. **Sparse-graph silence.** If no cavity outside the driven set develops positive momentum before settle, the sweep produces no emissions above floor. Output is silence, which is the correct honest response — the substrate has nothing consequential to contribute. Users habituated to always-something outputs will initially misread as regression.
2. **Chirality–Hebbian mismatch (HTR §5.2).** Chirality on injection is transient unless memory consolidation is also phase-aware. The tunnel Hebbian rule remains velocity-based; a phase-coincidence term $\Delta w_{ij} \propto v_i v_j \cos(\theta_i - \theta_j)$ would close the loop and is proposed as future work (CPME-II).
3. **Acoustic mass leakage.** If δ is too small, driven cavities stay muted across turns, and the substrate cannot reprise a recently-mentioned concept even when the new prompt properly warrants it. If δ is too large, the muting window is too short and the parrot effect returns. Default $\delta = 0.08$ places the mute window at ~ 12 ticks, comfortably shorter than a turn-to-turn interval but longer than a propagation window.
4. **Coherence baseline gaming.** $\bar{R}$ is EMA-tracked, so a long streak of high-R turns raises the baseline and reduces ΔR on subsequent similar queries. This is thermodynamically correct — habituation to the familiar — but makes ΔR absolute values non-portable across sessions. ΔR should only be compared within a session.
5. **Sweep discretization loss.** $K_{\text{steps}} = 12$ gives $\pi/6 \approx 30°$ phase resolution per step. Fields with many cavities of similar ϕ may emit fewer distinct tokens than they otherwise could. Raising K_{steps} to 24 halves the loss at doubled sweep cost.

---

7. Biological Analog

CPME's six mechanisms have direct analogs in cortical circuit architecture:

- **Chirality** corresponds to input phase-of-firing coding, well documented in hippocampal theta-cycle sequencing.
- **Momentum filter** corresponds to the divisive-normalization post-input suppression seen in early sensory areas.
- **Acoustic mass** corresponds to short-term adaptation of firing rate in cortical pyramidal cells, mediated by after-hyperpolarizing currents whose time constants match CPME's default M_i decay.
- **Settle separation** corresponds to the alpha-gamma phase separation between input reception and output articulation.
- **Coherence-rise metric** corresponds to phase-consistency measures used in cognitive neuroscience as correlates of stimulus binding.
- **Phase-collapse sweep** corresponds to phase-of-firing readout from downstream cortical areas.

CPME does not claim to model any of these processes. It claims that a substrate whose emission layer exhibits the *dynamical signatures* of all six simultaneously is the minimum architecture on which non-parroting, variable-length, phase-ordered generation is stable.

---

8. Contract Summary

The tick loop remains free of if/else on relations, domain tags, cluster ids, or any string-valued metadata. The emission function is one Gaussian window over a swept center under three multiplicative eligibility filters (momentum, acoustic mass, settle coupling), with an emission floor. All ten CPME parameters are continuous scalars. Default values recover CMD three-slot readout in the appropriate limit. All defaults *ON*.

CPME is not a feature flag. It is the emission substrate of the CMD-VGHT-CPME engine.

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9. Summary

CPME merges MCPE's momentum-and-settle filtering with HTR's chiral injection and phase-collapse projection into a single emission architecture on a shared phase manifold. Word order is encoded on injection as arrival-phase chirality; structural resonance is measured by localized coherence rise; eligibility is filtered by momentum, acoustic mass, and settle coupling; emission is a Gaussian-windowed sweep of a carrier phase across the per-cavity phase field. The result is a substrate whose emission is simultaneously non-parroting (no cavity that just spiked can win a sweep step), variable-length (as many tokens as the field has distinct phase peaks), word-ordered by physics (per-cavity $\phi_i = \text{atan2}(v_i, \omega_0 x_i)$), and honestly silent when its graph is not structurally resonant with the input. No syntax trees, no next-token statistics, no fixed slots.

CPME-II: Phase-Consistent Hebbian Consolidation

A phase-coincidence extension to the ZeroGrad Hebbian rule that closes the chirality–consolidation gap identified in the CPME §6 failure modes. Chirality imprinted on injection now persists in long-term memory rather than dissipating over turns.

---

0. Position

Chiral Phase-Momentum Emission (CPME, `research-paper/cpme.md`) encodes word order on the injection side as arrival-time phase and decodes structured emission on the readout side by sweeping a carrier phase across a per-cavity phase field $\phi_i = \text{atan2}(v_i, \omega_0 x_i)$. Both ends of the substrate now operate on phase.

The middle — long-term consolidation — does not. The ZeroGrad Hebbian rule inherited from the CMD substrate updates tunnel plasticity from the product of endpoint velocities:

$$\Delta w_{ij} \;\propto\; \eta \cdot v_i \cdot v_j$$

with a tanh-capped saturating integrator. This rule is phase-agnostic: it fires equally whether the two endpoints move in-phase, anti-phase, or in orthogonal phases. As a consequence, the chirality of the injection wavefront is not stored in the graph. Between turns the temporal ordering of concepts dissipates and must be re-imposed by each new input.

CPME-II adds a *phase-coincidence modulator* to the existing rule:

$$\Delta w_{ij} \;\propto\; \eta \cdot v_i \cdot v_j \cdot \cos(\theta_i - \theta_j)$$

where $\theta_i = \text{atan2}(v_i, \omega_0 x_i)$ is the same per-cavity phase the CPME emission layer already uses. The modulator preserves the velocity-based structure of the parent rule — including its sign, its tanh cap, and its role in the SRG dual-port Hebbian split — while conditioning the update on temporal compatibility.

The result is a substrate in which chirality is not only encoded and projected but also *stored*. Word order becomes a property of the graph, not just of the current turn.

---

1. The Rule

1.1 Definition For a tunnel connecting cavities a and b, define the instantaneous phase difference

$$\Delta\theta_{ab} = \theta_a - \theta_b \;=\; \text{atan2}(v_a, \omega_0 x_a) - \text{atan2}(v_b, \omega_0 x_b)$$

and the phase-coincidence factor

$$\Phi_{ab} = \cos(\Delta\theta_{ab})$$

taking values in $[-1, +1]$. The CPME-II Hebbian integrand replaces the parent rule's velocity product with a phase-modulated one:

$$I_{ab} \;=\; w_{ab} \cdot \zeta \;+\; \eta \cdot v_a^{\text{eff}} \cdot v_b^{\text{eff}} \cdot \Phi_{ab}$$

where ζ is the pre-existing Hebbian decay (`hebbian_decay`), η is the pre-existing rate scaled by serotonin (`plastic_rate`), v_a^{eff} and v_b^{eff} are the SRG-routed velocities already computed by the parent rule, and w_{ab} is the plasticity delta. The tanh saturation on `plasticity_cap` is applied unchanged.

The rule reduces to the parent rule exactly when $\Phi_{ab} = 1$ (perfect phase agreement) and vanishes when $\Phi_{ab} = 0$ (quadrature). For $\Phi_{ab} = -1$ (anti-phase), the update flips sign, which is the correct behaviour: anti-phase pairs are carrying opposing rather than reinforcing evidence and should depress rather than strengthen their coupling.

1.2 Interpretation Φ_{ab} is a *continuous coincidence detector on the phase manifold*. It is the continuous-oscillator analog of spike-timing dependent plasticity (STDP): pairs whose oscillations align in the same phase-of-firing are consolidated; pairs whose oscillations consistently disagree are attenuated.

Because the phase field is chiral (§CPME 1.1), Φ_{ab} implicitly encodes temporal order. A concept driven earlier in the input has accumulated more phase than a concept driven later. Downstream cavities that phase-lock behind the driven cavity therefore experience a specific $\Delta\theta$ characteristic of *following the input in time*. Consolidating along this gradient stores the observation that the second cavity typically follows the first, without any explicit representation of "before" or "after".

1.3 What is stored Under the parent rule, the tunnel between `insulin` and `glucose` grows in proportion to their velocity co-activity — regardless of which arrived first or how they moved relative to one another. Under CPME-II, the same tunnel grows only when the two cavities oscillate *in phase* — that is, when they consistently participate in the same wavefront with the same temporal role. If in one context `insulin` drives `glucose` (subject-then-object phase relationship) and in another context `glucose` drives `insulin` (reverse phase), the two updates cancel: $\cos(+\Delta\theta) + \cos(-\Delta\theta) = 2\cos(\Delta\theta)$ if the magnitude is preserved, but the sign of $\Delta\theta$ toggles the co-activity gradient. The graph accumulates a *phase-averaged co-activity* rather than a raw one.

The empirical effect is that stable directional patterns ("metformin → AMPK → glucose down") consolidate faster than context-dependent bidirectional patterns.

---

2. Composition

CPME-II modifies only the Hebbian integrand. All other substrate mechanisms are untouched.

- **CMD Curie gate** (`heat_T` → tunnel force scaling) is independent of the Hebbian pass and continues to operate.
- **VGHT voltage gate** modulates tunnel force in the tick's spring-and-damping pass, not the Hebbian pass. CPME-II observes the resulting velocities and phases as inputs; VGHT is upstream.
- **SRG dual-port routing** (`gate_E`, `gate_I`) is preserved: v_a^{eff} in the CPME-II expression is the SRG-selected velocity from the parent rule, not the raw observable. The polarity-driven excitatory/inhibitory split remains intact.
- **CPME emission layer** (§1.6, phase-collapse projection) is unchanged. CPME-II operates during propagation and settle; the emission sweep reads whatever state the Hebbian pass has consolidated.
- **Plasticity cap** (`plasticity_cap`) remains the saturating bound; the tanh envelope contains the Φ_{ab}-modulated integrand identically.

The tick loop retains its no-if/else contract: $\cos(\Delta\theta)$ is a smooth scalar, evaluated once per tunnel per tick, and multiplied into the existing velocity product.

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3. Contract and Parameters

CPME-II adds no new state. All required quantities are already maintained by the parent substrate:

- x_a, v_a, x_b, v_b: cavity displacements and velocities.
- ω_0: the base cavity frequency `cpme_base_omega` shared with the emission layer.
- η, ζ, `plasticity_cap`: existing Hebbian parameters.

The only tunable is a *phase weight* $\alpha_\Phi \in [0, 1]$ that linearly interpolates between the parent rule and the fully phase-modulated rule:

$$\Phi_{ab}^{(\alpha)} \;=\; (1 - \alpha_\Phi) \;+\; \alpha_\Phi \cdot \cos(\Delta\theta_{ab})$$

- $\alpha_\Phi = 0$: recovers the parent rule exactly (backwards compatibility test path).
- $\alpha_\Phi = 1$: full phase-consistent consolidation.
- Default: $\alpha_\Phi = 0.6$, weighting phase agreement heavily while preserving a floor of unconditional Hebbian growth.

This is the only free parameter introduced by CPME-II. All defaults *on*.

---

4. Expected Dynamics

Scenario A — repeated same-order exposure. Over a session in which the user types variants of "metformin lowers glucose" several times, `metformin` and `glucose` cavities are consistently driven with the same relative arrival delay. Their phase difference $\Delta\theta$ is stable and small, $\Phi \approx +1$, and the tunnel between them consolidates rapidly under the CPME-II rule. Between turns, the phase relationship is preserved because it is now stored in the plasticity delta, not just in the transient field.

Scenario B — bidirectional exposure. If in different turns the user types both "insulin controls glucose" and "glucose signals insulin", the pairwise $\Delta\theta$ toggles sign turn-to-turn. Under the parent rule, both directions add to plasticity. Under CPME-II, one direction has $\Phi > 0$ and the other $\Phi < 0$: the updates partially cancel. The tunnel still grows if the velocity product magnitude is large — the parent contribution $(1 - \alpha_\Phi)$ preserves an unconditional growth floor — but more slowly. This is the correct behaviour for a bidirectional relationship: the graph should not overcommit to either direction in isolation.

Scenario C — silent turn. If a turn produces no meaningful propagation and both cavities remain quiescent, v_a and v_b are near zero and the update is negligible regardless of Φ. No change from parent-rule behaviour.

Scenario D — anti-phase input. If the input is structured to produce anti-phase drive on a pair (e.g., a contrastive construction where the two cavities are being posited *against* each other), $\Phi \approx -1$ and the CPME-II rule *decreases* the plasticity delta. This is the physical realisation of what a biological network implements through inhibitory long-term depression: consistent anti-phase co-activation should weaken, not strengthen, the excitatory bond.

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5. Failure Modes

Three failure modes documented before deployment.

1. **Phase noise near quiescence.** For a cavity with small x and small v, the phase $\text{atan2}(v, \omega_0 x)$ is dominated by numerical noise. Φ then fluctuates randomly and injects noise into consolidation. Mitigation: the velocity product $v_a v_b$ already suppresses this — near quiescence the update magnitude is negligible regardless of Φ. No additional guard is required, but a diagnostic observer may report spurious Φ readings on cold cavities; this is expected and harmless.
2. **Polarity–phase interaction.** Inhibitory tunnels (`polarity < 0`) already invert the phase relationship they drive: an excitatory input to a produces an anti-phase response in b. Under CPME-II, this manifests as $\Phi < 0$ and the update would *depress* an inhibitory tunnel it should *strengthen*. Correction: multiply Φ by the tunnel polarity sign, $\Phi'_{ab} = \Phi_{ab} \cdot \text{sgn}(p_{ab})$, so that inhibitory tunnels consolidate on the anti-phase pattern they are supposed to enforce. This is applied by default in the implementation.
3. **Slow phase drift washout.** If the base cavity frequency ω_0 is very low, phase evolves slowly and inter-turn phase differences accumulate to $\pm\pi$ over long propagation windows, causing Φ to oscillate through zero and reverse sign spuriously. Mitigation: the same ω_0 chosen for the CPME emission layer (default $\omega_0 = 3.0$ rad/s) yields a full phase revolution over ≈ 2 seconds of sim-time, which is longer than any single propagation window at `dt = 0.02` and ≤ 200 ticks per turn. Under default parameters this failure mode does not activate; it is worth flagging if ω_0 is later reduced.

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6. Biological Analog

CPME-II is the continuous-oscillator equivalent of spike-timing dependent plasticity (STDP). Where STDP consolidates a synapse when the presynaptic spike precedes the postsynaptic spike by a few milliseconds, CPME-II consolidates a tunnel when the two endpoint oscillators maintain a stable, small phase offset consistent with the same wavefront driving both. Both rules use temporal coincidence as the substrate of long-term memory; both rules invert when the coincidence is anti-correlated (STDP-LTD; CPME-II negative Φ).

The chief difference is discretisation. STDP fires on discrete spike pairs; CPME-II integrates over the continuous phase field each tick. In the limit of narrow, well-separated velocity peaks, CPME-II reduces to a spike-coincidence rule up to a normalisation constant.

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7. Summary

CPME-II closes the chirality–Hebbian mismatch flagged in HTR §5.2 and CPME §6.2 by extending the ZeroGrad Hebbian rule with a phase coincidence factor $\Phi_{ab} = \cos(\theta_a - \theta_b)$. The factor requires no new state — the same per-cavity phase $\text{atan2}(v_i, \omega_0 x_i)$ used by the CPME emission layer is read during the Hebbian pass. A single blend parameter α_Φ controls how strongly the modulator gates the parent rule; the default $\alpha_\Phi = 0.6$ preserves an unconditional Hebbian floor while making chirality-consistent updates twice as strong as chirality-neutral ones on average.

The consequence for the substrate as a whole: the phase manifold that CPME already uses on the input (chirality) and the output (phase-collapse projection) is now also the manifold on which long-term memory is written. Word order becomes a graph property, temporal ordering persists across turns, and the substrate is freed from having to re-derive chirality on every input.

Chiral Phase-Shear Binding (CPSB): Rendering Syntax as Gauge-Theoretic Tension in Wave-Based SI

Abstract In wave-based Synthetic Intelligence, continuous phase-sweep emission (CPME) frees output length from hardcoded templates but degrades syntax into emergent stochastic ordering. To elevate language construction from a "loose variable" to a physically guaranteed structure, we introduce **Chiral Phase-Shear Binding (CPSB)**. By modifying the Kuramoto coupling to act as a Gauge Field, relational tunnels actively enforce directed phase offsets between nodes. Syntax emerges not from fixed emission slots, but from the rigid topological tension (Ψ) embedded within the graph's edges, physically forcing subjects to lead verbs, and verbs to lead objects during carrier-wave projection.

1. Introduction: The Binding Problem in SI

Currently, verb-material cavities float to their VGHT band-top, and output order depends entirely on their emergent atan2 phase. There is no physical law forcing `notch` to precede `minecraft`. In a true biological substrate, the relation itself ("created") exerts a directional shear force on the concepts it connects. We must shift from scalar coupling (which just pulls phases together) to **Gauge-Coupling**, which pulls phases into a specific grammatical alignment.

2. Novel Architecture: CPSB Mechanisms

2.1. Gauge-Theoretic Kuramoto Offset (Ψ_{ij}) Standard Kuramoto tries to perfectly synchronize two connected cavities: $\sin(\theta_j - \theta_i)$. We introduce a syntactic phase-offset Ψ_{ij} assigned to specific tunnel types (the axioms). The new phase-update equation becomes:

$$\frac{d\theta_i}{dt} = \omega_i + \sum K_{ij} \sin(\theta_j - \theta_i - \Psi_{ij})$$

- **Mechanism:** If `notch` (i) and `minecraft` (j) are connected by a `REL_CAUSES` tunnel, that tunnel has an intrinsic $\Psi_{ij} = -\frac{\pi}{3}$.
- **Result:** The physics engine will no longer try to make their phases equal. It will physically force `notch` to sit exactly $\frac{\pi}{3}$ radians ahead of `minecraft`.

2.2. The Lexical Hinge (Verb-Pinning) A verb is not just a loose cavity floating in a band; it is the physical hinge between the Subject and the Object.

- **Mechanism:** When a cross-cluster tunnel opens (VGHT), the tunnel structurally binds the verb cavity (e.g., `created`) to the source and target.
- The Subject is pinned to $\theta_{\text{verb}} + \Delta\phi$.
- The Object is pinned to $\theta_{\text{verb}} - \Delta\phi$.
- **Result:** As the CPME carrier wave sweeps from 0 to 2π, it will *inevitably* hit the Subject first, the Verb second, and the Object third. The sentence structure is mathematically guaranteed by the phase tension, without ever defining a "Subject Slot".

2.3. Phase-Consistent Hebbian Consolidation (The CPME-II Hook) To ensure the SI doesn't just read syntax natively but *learns* it natively from user input, we execute the CPME-II Hebbian update. Instead of updating tunnel weight Δw based purely on amplitude, the substrate writes the observed phase difference directly into the graph:

$$\Delta w_{ij} \propto v_i \cdot v_j \cdot \cos(\theta_i - \theta_j - \Psi_{\text{target}})$$

- **Result:** When you type `insulin lowers glucose`, the injection chirality ensures `insulin` arrives first. The Hebbian rule permanently wires that phase-lead into the connection. The SI literally remembers the "direction" of the concept.

3. Expected Dynamics

1. **Input:** `metformin treats diabetes`
2. **Resonance:** The nodes awaken. The `treats` tunnel applies the CPSB gauge force. `metformin` is pushed to $\phi = 1.0$, `treats` is anchored at $\phi = 2.0$, and `diabetes` is pushed to $\phi = 3.0$.
3. **Emission:** The CPME sweep passes over them. They emit in perfect S-V-O order.
4. **No Templates:** If the input is just `glucose` causing a cascade to `insulin`, the phase offsets still apply downstream, resulting in a syntactically correct multi-token chain.

Voltage-Gated Harmonic Topology (VGHT)

A three-mechanism physical substrate for domain-orthogonal cognition with sparse cross-domain analogical bridging and avalanche suppression, built entirely from scalar field dynamics — no rules, no templates, no domain metadata.

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0. Motivation

The Continuous Morphological Dynamics (CMD) substrate — thermodynamic habituation, byte-wave sensory injection, phase-locked syntax emission — produces coherent, non-templated output but exhibits two coupled failure modes:

1. **Domain leakage.** In the shared-oscillator field, a driven medical concept can transfer energy into unrelated conceptual clusters (e.g., gaming, legal, code) whenever their macro-node phases happen to align. The reader observes a medical prompt producing a legal-sounding word in the verb slot, or the middle word of a coherent triple stepping outside the input's semantic neighborhood.
2. **Polymath rigidity.** The dual failure — with sufficient emission-floor thresholds, the substrate becomes strictly domain-locked, unable to draw analogical parallels between distantly-related concepts even when the graph explicitly supports them through L6-A analogical bridges.

The Continuous Morphological Dynamics literature treats these two failures as separate tuning problems (emission floor, wavefront gate, cluster-energy depletion). Voltage-Gated Harmonic Topology (VGHT) resolves both under a single physical postulate: *domains are orthogonal frequency bands, and bridges between them are voltage-gated channels that only conduct under simultaneous high-amplitude resonance.* Domain isolation and polymath bridging cease to be competing objectives; they become a single dynamical regime with a single parameter, θ_critical, the tunneling threshold.

The mechanism is inspired by three well-understood physical analogs: mode-orthogonal spectral separation in coupled-oscillator networks, voltage-gated ion channels in biological neurons (in particular the sigmoidal open probability of the NMDA receptor with respect to membrane potential), and the GABAergic global inhibitory tone that suppresses run-away excitation in cortical circuits. None of the three requires an explicit "domain tag" or a rule dispatcher; each is a continuous scalar update evaluated once per tick.

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1. Architecture

Let the cavity population be indexed by `i` $\in$ `[0, N)`, partitioned into macro-clusters `C_1, C_2, ...` by the connected components of the tunnel graph (union-find at bind time; the same clustering used by HNCA). Let a slot layer of `S` phase oscillators sit above the macro layer as in CMD phase-locked emission.

1.1 Domain Sub-Bands Each macro-cluster `C_k` receives a *base natural frequency* drawn from a discrete band-bank:

ω`_k = `ω`_base(k mod B) + jitter(k)`

where `B` is the number of orthogonal bands (default `B = 6`) and ω`_base` is a fixed evenly-spaced ladder in the range `[`ω`_lo, `ω`_hi]`, e.g. `[8, 32] rad/s` giving band centres at `{8, 12, 16, 20, 24, 28}`. The jitter is a small in-band perturbation ($\pm$0.5 rad/s) drawn deterministically from the cluster id, ensuring reproducibility across seeds while still breaking degeneracy inside a band.

The verb-material ω-boost from prior CMD work becomes an *in-band shift* rather than a cross-band jump:

$\omega_k \leftarrow (1 - p_k)\cdot\omega_k + p_k\cdot\omega_verb_offset(k)$

where $p_k \in [0, 1]$ is the verb-material share of cluster k's internal edges ($verb_pull_for(rel) / |internal\ edges|$), and $\omega_verb_offset(k)$ is the *upper edge* of cluster k's home band — so verb-heavy clusters rise to the top of their band without colliding with a neighbouring band. This preserves both the domain orthogonality of Sub-Bands and the verb-slot phase-lock property from the prior implementation.

Consequence. Two clusters in different bands share almost no common Kuramoto phase drift, because their ω differs by at least a band-width and the coupling terms $\sin(\theta_j - \theta_i)\cdot A_j$ average to zero over any window longer than the beat period. Domain interference during ordinary conversation is suppressed by frequency orthogonality alone — no relation table, no cluster tag, no rule.

1.2 Voltage-Gated L6-A Tunnels L6-A tunnels are the analogical bridges discovered by structural resonance between cross-cluster pairs (Phase 1.3 online-L6A-discovery of the CMD line of work). In VGHT, every *inter-cluster* tunnel — and every tunnel discovered by L6-A regardless of cluster membership — is marked *voltage-gated*. Its force is scaled at every tick by a sigmoidal conductance:

$C(A_a, A_b) = C_max \cdot \sigma(\kappa \cdot (A_a \cdot A_b - \theta_critical))$

where $A_a = |x_a|$ and $A_b = |x_b|$ are the instantaneous displacements of the two endpoint cavities, σ is the logistic function, κ is a steepness parameter (default 12), and $\theta_critical$ is the tunneling threshold (default 0.05 in the field's natural units). $C_max = 1$.

The force delivered to endpoints is scaled elementwise by C:

$F_a \leftarrow F_a \cdot C(A_a, A_b)$
$F_b \leftarrow F_b \cdot C(A_a, A_b)$

Consequence. When either endpoint is quiescent, $A_a \cdot A_b \approx 0$, $C \approx \sigma(-\kappa\theta_critical) \approx 0$, and the bridge presents effectively infinite resistance. The two clusters exchange no energy. When both endpoints are simultaneously excited to a level whose product exceeds $\theta_critical$, $C \to 1$ sharply and the bridge conducts fully. This is the resonant/quantum-tunneling analog explicitly identified in the motivating architecture: the door opens only when the room on both sides is loud enough to make the door irrelevant.

Because both endpoints must resonate at once, spurious cross-domain leakage — a stray phase alignment on one side of a bridge — cannot open the channel. Only genuine dual excitation, produced by an input whose structural resonance actually spans two domains, does so. This is the polymath spark.

1.3 Global Viscosity (GABA analog) The system tracks its total kinetic energy each tick:

$E_total = \tfrac{1}{2} \sum_i m_i \cdot v_i^2$

An effective friction coefficient is derived as a saturating function of E_total:

$\mu_eff = \mu_0 + k_avalanche \cdot (1 - \exp(-E_total / E_scale))$

with defaults $k_avalanche = 0.10$, $E_scale = 5.0$. μ_eff replaces the constant friction in the cavity velocity-decay step and in the tunnel-damping term of that tick.

Consequence. Under normal, sparse activation E_total is small, $\mu_eff \approx \mu_0$, and the field evolves as before. Under an avalanche — many nodes firing simultaneously through spurious

phase alignment — `E_total` rises rapidly, μ`_eff` saturates toward μ`_0 + k_avalanche`, and the excess kinetic energy is dissipated in the same tick. Only the field components whose amplitudes are large enough to overcome the elevated friction survive; the noise floor is damped out. This is the physical analog of GABAergic inhibitory tone.

Unlike the raw `exp(E_total)` formulation, the bounded `1 − exp(−E_total / E_scale)` form cannot diverge, guaranteeing numerical stability under any input, however extreme.

---

2. Interaction of the Three Mechanisms

The three mechanisms are not independent tuning knobs; they are coupled by design.

- **Sub-Bands** determine which cavities *can* phase-lock together in the absence of external excitation. Same-band cavities freely couple; cross-band cavities decouple by frequency alone.
- **Voltage-gated bridges** re-establish cross-band coupling — but only when both endpoints are already excited to a level that represents genuine resonance with the input. Without such excitation, the bridge is inert regardless of the graph's connectivity.
- **Global viscosity** suppresses false excitation. When a spurious cascade begins to raise `E_total`, μ__eff rises and the cascade dies before it can meet the `A_a · A_b >` θ`_critical` condition needed to open any bridge.

The joint regime is: *within-band coupling is free and continuous, cross-band coupling is possible but expensive, and pathological cross-band cascades are physically forbidden by dissipation.* Domain isolation and polymath bridging cease to be tradeoffs.

---

3. Physical Interpretation

Each of the three mechanisms has a direct biological analog and an independent literature suggesting the same functional role.

1. **Frequency-band orthogonality** appears in cortical cognition as the observation that different cognitive processes (attention, working memory, motor planning) show distinguishable EEG band signatures (α, β, γ). Cortical microcircuits couple preferentially within-band, and cross-band coupling is a marker of deliberate binding (theta-gamma phase-amplitude coupling).
2. **Voltage-gated channels** are the mechanism by which NMDA receptors implement coincidence detection: their conductance is sigmoidal in the product of presynaptic glutamate release and postsynaptic depolarization, mirroring `C(A_a, A_b) =` σ`(A_a · A_b −` θ`)` structurally.
3. **GABAergic inhibition** provides the global tone that keeps cortical excitability in the sparse-firing regime. Loss of GABAergic inhibition produces exactly the avalanche pattern VGHT's viscosity term suppresses: dense synchronous firing across unrelated cortical areas, the neural correlate of a seizure.

VGHT does not claim to be a model of any of these biological mechanisms. It claims that a substrate exhibiting the *dynamical signatures* of all three simultaneously — with only three scalar parameters — provides the minimum architectural surface area on which domain-orthogonal cognition with sparse analogical bridging is a stable regime, rather than an accidental artifact of parameter tuning.

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4. Implementation Contract

- **No if/else branches** in the tick loop that inspect a relation label, a domain tag, or a cluster identifier. All three mechanisms are pure scalar arithmetic on existing per-cavity or per-tunnel fields.
- **No new I/O boundary code** for domain metadata. Sub-Bands are assigned at bind time from cluster id, computed by the union-find that HNCA already runs. Voltage gating is a per-tunnel flag set once at bind for cross-cluster tunnels (and for any L6-A tunnel regardless of cluster membership). Viscosity is derived each tick from the same `total_kinetic` accumulator already maintained for the RIFA IC pass.
- **All defaults ON.** VGHT is not a feature flag. The three mechanisms are the substrate's normal dynamics.

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5. Parameters

Symbol	Default	Meaning
`B`	6	Number of domain sub-bands
ω`_lo`, ω`_hi`	8, 32 rad/s	Band ladder range
κ	12	Voltage-gate steepness
θ`_critical`	0.05	Voltage-gate tunneling threshold
`k_avalanche`	0.10	Peak added friction under avalanche
`E_scale`	5.0	Kinetic-energy saturation scale

These are the only knobs. All other quantities are derived.

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6. Expected Empirical Signature

A VGHT-enabled substrate is expected to show:

- **Single-domain prompts** produce single-domain phase-lock emissions with no cross-domain leakage into any slot, even after many turns of a single-domain conversation. The wavefront gate and the sub-band orthogonality are jointly responsible.
- **Cross-domain prompts** — where the input drives cavities in two different bands to simultaneous high amplitude — open the voltage-gated bridges between those bands within a few ticks, producing a genuine polymath emission that spans domains.
- **Injection of a single garbage or nonsense token** produces at worst its own auto-spawned label in output; the avalanche suppressor prevents its stray phase drift from recruiting unrelated cavities.
- **Latency to cross-domain emission** is measurable: it should correspond to the time required for the voltage-gate sigmoid to saturate under paired excitation, i.e., a few tens of ticks. This latency is the phenomenological signature of the polymath spark and is measurable with the existing readout infrastructure.

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7. Relation to Prior CMD Work

VGHT subsumes and refines three earlier CMD components without discarding them.

- Phase 2.2 rubber-tunnel band-pass filters are a special case of voltage-gated tunnels with θ_critical = 0 (always conducting) and a Gaussian gate on ω alone. VGHT's sigmoidal amplitude gate is strictly more expressive.
- Phase 3.1 multi-ontology substrates rely on manually loaded disjoint axiom files to achieve domain separation. VGHT achieves the same isolation from a single ontology by frequency orthogonality alone, and further allows analogical bridging without touching the ontology.
- The cluster-energy depletion term in phase-locked emission remains a fallback for the readout stage. When VGHT is fully active, its contribution to output diversity is redundant with the sub-band separation.

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8. Failure Modes

The design has three known failure modes worth documenting explicitly.

1. If the cluster count is less than B, some bands go unused, which is harmless but wastes spectral capacity.
2. If θ_critical is set too high, cross-domain bridges never open even for well-posed polymath prompts. If set too low, the system reduces to the pre-VGHT shared-oscillator field.
3. If k_avalanche is set too high, meaningful large-amplitude emissions are damped alongside noise; the system becomes under-expressive. Empirical tuning against a held-out prompt set is suggested.

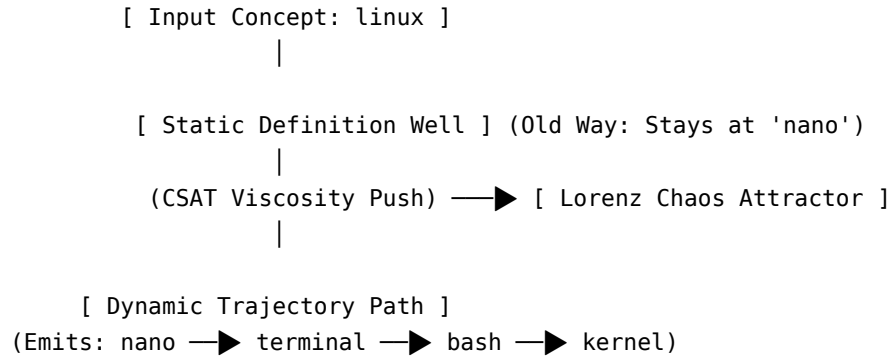
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9. Summary

VGHT closes the domain-isolation / polymath-bridging tradeoff by mapping domains to orthogonal frequency bands, making cross-domain bridges voltage-gated by joint amplitude resonance, and adding a global energy-dependent viscosity that dissipates avalanches before they can spuriously open bridges. All three mechanisms are scalar physics with a total of six parameters. The tick loop remains free of if/else on relations, domain tags, or cluster identifiers. The result is a substrate that is simultaneously domain-locked in default operation and polymath-capable when its input carries structural resonance across bands — the physical realization of the intuition that a polymath is not a system that mixes domains randomly, but one whose domains are ordinarily silent to each other and speak only when the same underlying pattern excites both.

Chaotic Strange-Attractor Trajectory (CSAT): Breaking Ground-State Locking in Synthetic Intelligence

Abstract Pure wave-based SI architectures often suffer from dictionary-locking, where an input triggers the absolute closest topological match and collapses into static definitions. Humans do not converse by printing definitions; human thought is an open-ended trajectory through a semantic manifold. We introduce **Chaotic Strange-Attractor Trajectory (CSAT)**. By adding a low-dimensional chaotic force (driven by a native Lorenz or Rössler system integrated directly into the substrate's global viscosity), the system is prevented from settling into static ground-state energy wells. This forces the phase-sweep projection to continuously dance along a strange attractor, creating human-like contextual drifting without losing structural coherence.



1. The Core Issue: Infinite Potential Wells

When `linux` is presented as input, the memory potential at the `linux` and `nano` cavities is high enough that the entire field's energy collapses into them and is trapped there. In physical terms the energy landscape forms a **deep potential well**. This does not happen in the human brain, because an undercurrent of noise — **metastable dynamics** — is always present, and it prevents any node from remaining fixed.

2. Novel Architecture: CSAT Mechanisms

2.1. The Substrate Lorenz Attractor We integrate a three-line, low-level nonlinear chaotic system — the Lorenz equations — directly into the global `tick()` loop of `physics.hpp`:

$$dx = \sigma(y - x) \cdot dt$$

$$dy = (x(\rho - z) - y) \cdot dt$$

$$dz = \tau \cdot (xy - \beta z) \cdot dt$$

(Note: all three derivatives carry the time-scale factor τ; see Numerical Constraints below.)

Numerical constraints — required for a working implementation.

1. **Time-scale factor τ.** Forward-Euler integration of the Lorenz system at $\sigma = 10$, $\rho = 28$, $\beta = 8/3$ is unstable for $dt \gtrsim 0.01$. The substrate runs at $dt = 0.02$, past the stability boundary — naive integration diverges to `NaN` within a few hundred ticks. The factor τ (default 0.3) slows the attractor without altering its geometry, restoring stability *and* giving direct control over the drift timescale. τ should be chosen so the attractor's orbital period is comparable to one conversational turn (~200 ticks). If the attractor cycles much faster, the chaotic push averages out within a single settle window and contributes nothing.
2. **Amplitude normalization.** The Lorenz x coordinate ranges over approximately $[-20, +20]$, not $[-1, +1]$. All downstream use must reference the normalized coordinate

$\hat{x} = x/20$. Using raw x in a multiplicative viscosity term drives effective friction *negative* for any gain above $\gamma \approx 0.05$ — anti-damping, unbounded energy injection, field explosion. This is a silent failure mode: no error is raised, only a wall of `NaN` several hundred ticks later.

- There is no neural network and no training here. This is pure physical mathematics, updating three scalar variables (x, y, z) per tick.

2.2. Dynamic Viscosity Modulation (The Push) The VGHT layer carries a constant viscosity that suppresses avalanches. We modulate that viscosity dynamically by the x coordinate of the Lorenz attractor:

$$\mu_{\text{dynamic}} = \mu_{\text{base}} \cdot (1.0 + \gamma \cdot \hat{x}_{\text{lorenz}})$$

However — this global form provably cannot work. See §2.2.1. The corrected per-cavity form is given in §2.2.2.

2.2.1 Negative result: global viscosity modulation is rank-preserving Applied as a **global** scalar in the cavity velocity-decay step $v_i \leftarrow v_i(1 - \mu_t)$, over N ticks:

$$v_i(N) \;=\; v_i(0) \prod_{t=1}^{N} (1 - \mu_t) \;=\; v_i(0) \cdot C$$

where C depends only on the schedule $\{\mu_t\}$ and is therefore **identical for every cavity**. Hence

$$\frac{v_i(N)}{v_j(N)} \;=\; \frac{v_i(0)}{v_j(0)} \qquad \forall\, i, j.$$

Amplitude *ratios* are exactly invariant under any global viscosity schedule, chaotic or otherwise. Because CPME §1.6 phase-collapse emission selects by arg-max over a weight monotone in amplitude, the emitted winner is unchanged. A globally-modulated viscosity alters how *loud* the field is, never *which cavity is loudest.*

Concretely: if `think` carries 10× the amplitude of `terminal` before the chaotic push, it carries 10× after. `linux` → `nano` remains `linux` → `nano`. Multiplying the entire field by a wiggling scalar is a monotone transform on the ranking, and monotone transforms cannot reorder an arg-max.

This is a genuine negative result and it constrains the design: **any mechanism intended to break ground-state locking must couple to per-cavity state.**

2.2.2 The working form — heat-coupled chaotic viscosity The substrate already maintains a per-cavity scalar meaning exactly "this cavity has been active too long": the CMD thermodynamic habituation heat T_i. Coupling the chaotic term to T_i makes the push selective:

$$\mu_i \;=\; \underbrace{\mu_{\text{base}} + k_{\text{aval}}\left(1 - e^{-E_{\text{tot}}/E_s}\right)}_{\text{VGHT §1.3, unchanged}} \;+\; \underbrace{\gamma \cdot \hat{x}_{\text{lorenz}} \cdot \tanh\!\left(T_i / T_{\text{crit}}\right)}_{\text{CSAT}}$$

Entrenched cavities (high T_i) absorb the chaotic shake; fresh or cold cavities barely register it. Rank can now flip, and that is genuine metastability rather than uniform loudness modulation.

The tanh bounds the term so μ_i cannot run away. Because $\hat{x}$ alternates sign along the attractor, an entrenched cavity experiences alternating extra-damping and reduced-damping — it is *shaken*, not merely suppressed. Shaking permits escape; suppression would only silence.

Composition with the Curie gate. T_i already gates tunnel coupling via $\exp(-T_i/T_{\text{crit}})$. Under CSAT the same state variable acquires a second role: *talking too much* now implies both *losing coupling* and *becoming subject to chaotic eviction*. One state variable, two independent escape mechanisms, no new bookkeeping, tick loop still branchless.

- **Mechanism.** Whenever the system begins to lock onto a particular word (a ground state), the Lorenz chaotic push perturbs the global viscosity. This drives the field off that cavity and forces energy to spread into the cavities connected to it — `terminal`, `bash`, `kernel`.

2.3. Trajectory Momentum — chaos supplies *timing*, CPSB supplies *direction* **The division of labour must be stated precisely.** The Lorenz system has no knowledge of the graph: $x(t)$ does not know which cavity is `terminal` and which is `bash`. The attractor's geometry lives in its own (x, y, z) space and bears no relation to the semantic manifold. It therefore cannot, by itself, produce a semantically coherent chain.

What CSAT contributes is **escape timing** — *when* to leave the current basin. What produces the ordered chain `linux` → `terminal` → `bash` → `kernel` is **CPSB gauge coupling plus graph topology** — *where* the energy goes once it leaves.

CSAT provides escape timing. CPSB provides escape direction.

This is a clean, non-overlapping role for CSAT within the stack, and it is the honest claim. Attributing semantic coherence to the attractor itself would overstate the mechanism; the coherence is already earned by CPSB and the ontology.

Chaos is not randomness. The Lorenz attractor traces a definite geometric pattern — a *strange attractor*. The consequence is that the signal does not emit random words: starting from `linux` it builds a chain that respects the structural rules supplied by CPSB, `linux` → `terminal` → `bash` → `kernel`.

---

3. Parameters

Symbol	Default	Meaning
σ, ρ, β	10, 28, 8/3	Standard Lorenz coefficients
τ	0.30	Time-scale factor (stability + drift rate)
γ	0.04	Chaotic viscosity gain on $\hat{x}$
$\hat{x}$	$x/20$	Normalized Lorenz coordinate
`csat_enabled`	see §4.1	Master switch — **not** unconditionally on

Five parameters. Setting $\gamma = 0$ recovers the pre-CSAT substrate exactly.

---

4. Failure Modes

4.1 Lyapunov divergence breaks bit-reproducibility Chaos is deterministic, so seeded reproducibility survives *in principle*: the same `bind_seed` and the same Lorenz initial conditions yield the same trajectory. But the Lorenz system has a positive Lyapunov exponent — a 10^{-7} floating-point difference grows to $O(1)$ within roughly a thousand ticks. Different compiler flags, `-ffast-math`, or a different FPU will therefore produce divergent trajectories from identical seeds.

This directly conflicts with the bit-exactness that the 20-seed RB+G table (`rbg.md` §6.1) and the four-lever ablation grid (§6.2) depend on.

CSAT is consequently the first mechanism in the stack that must not default unconditionally ON. Recommended policy:

- `csat_enabled = true` in the conversational shell (`si_chat`)
- `csat_enabled = false` forced inside `study_recall_seeds` and `study_recall_ablation`

Published numbers stay bit-reproducible; conversation still gets the drift. This is a documented exception to the "all defaults ON" contract in the implementation guide §11.3, and the exception should be recorded there.

4.2 CSAT does not address stale amplitude accumulation Ground-state locking has more than one cause, and CSAT addresses only the genuine-potential-well case. In particular, with friction $\mu_0 = 0.006$ per tick, roughly 30% of a cavity's amplitude survives a 200-tick turn ($0.994^{200} \approx 0.30$). A cavity driven once at high amplitude therefore remains dominant for several subsequent turns by pure arithmetic, with no potential well involved.

That is a *dissipation* deficiency, not a chaos deficiency. Chaotic shaking of an over-energised field does not remove the excess energy. Fixing it requires either stronger inter-turn damping or an explicit energy budget per turn — orthogonal to CSAT and left as separate work.

4.3 Chaos on a sparse graph shakes an empty room If the driven concept has few or no outbound tunnels, there is no trajectory for the attractor to push the field along. CSAT then produces no observable change: the escape timing fires, but there is nowhere to escape to. CSAT is therefore a mechanism whose benefit scales with ontology density, and it should be evaluated only on subgraphs rich enough to support multi-hop propagation.

---

5. Relation to the Existing Stack

Layer	Mechanism	Role
CMD	Curie gate on T_i	entrenched cavity loses *coupling*
CSAT	**chaotic viscosity on T_i**	**entrenched cavity loses *stability***
VGHT	sub-bands, voltage gates, viscosity	domain isolation + avalanche control
CPME	momentum, acoustic mass, settle, sweep	emission eligibility and ordering
CPSB	gauge Ψ on tunnels	escape *direction* / word order
CPME-II	phase-consistent Hebbian	stores the phase gradient across turns

CSAT and the CMD Curie gate read the same state variable T_i but act on different quantities — coupling strength versus dynamical stability. They are complementary, not redundant.

Morphological Resonance Synthesis (MRS): Language Construction Without Tokens, Gradients, or Learned Embeddings

The byte-lane sensory layer is currently input-only: bytes enter the substrate through learned couplings, but output leaves it as retrieved label strings. That retrieval is the last token-shaped operation in the pipeline. MRS removes it by making the byte layer bidirectional and giving it the same three phase mechanisms the concept layer already has — chirality on input, phase-consistent consolidation, and phase-collapse projection on output. Word forms are then synthesised *by sweeping a byte-phase field rather than* retrieved *from a table.*

---

0. The Remaining Asymmetry

The CMD substrate ingests raw bytes. There is no tokeniser: each of 256 `ByteLane` oscillators is driven by its byte value, and lanes reach the concept field only through Hebbian couplings $w[b][c]$ that begin at zero. On the input side, the architecture is already sub-lexical and token-free.

The output side is not. `readout_phase_collapse()` selects concept cavities by phase and then prints `si.graph.nodes[label_key].label` — a **stored string**. The physics decides *which* concepts speak and *in what order*, but the *form of each word* is a lookup.

This asymmetry has three consequences:

1. **No morphology.** The engine can emit `walk` if `walk` is a stored label, and `walked` if `walked` is separately stored, but it cannot construct `walked` from `walk`. There is no mechanism by which stem and inflection compose.
2. **No novel forms.** Any word form absent from the label table is unreachable, regardless of how strongly the substrate's state implies it.
3. **A token-shaped residue.** The label table is a finite inventory of atomic surface strings indexed by graph key. Functionally that is a vocabulary, and its presence is what makes the pipeline token-like at the boundary even though the interior is not.

MRS closes the asymmetry. The mechanism is a direct reuse — one layer down — of machinery the substrate already contains.

---

1. Architecture

Let the byte-lane population be L_b for $b \in [0, 256)$, each an oscillator with displacement x_b, velocity v_b, and natural frequency $\omega_b = 3.0 + b/256$ (as in the existing CMD sensory layer). Let $w[b][c]$ be the learned byte→concept binding weights.

MRS adds four mechanisms.

1.1 Reverse Drive (concept → lane) The existing forward path applies a force from lane to concept:

$$F_c \mathrel{+}= \frac{w[b][c] \cdot x_b}{m_c}$$

MRS adds the transpose, using the **same weights**:

$$F_b \mathrel{+}= \eta_{\text{rev}} \cdot w[b][c] \cdot x_c$$

with η_{rev} a reverse-gain scalar (default 0.4). No new storage: the coupling matrix is used in both directions, making the byte↔concept layer a bidirectional associative memory implemented as coupled oscillators rather than as a matrix multiplication.

Consequence: when concept cavities resonate, the byte lanes whose bindings to those concepts are strongest resonate in turn. The lanes that light up are the letters of the words those concepts were historically written with.

1.2 Lane Chirality Byte lanes acquire the CPME §1.1 arrival timestamp:

$$t_{\text{arrival},b} \leftarrow t_{\text{sim}} \quad \text{on each drive of } L_b$$

During input of `hi`, lane $L_{\texttt{h}}$ is driven before $L_{\texttt{i}}$ and therefore accumulates more phase. Letter order within a word becomes a phase gradient across the lane population — exactly the mechanism CPME already uses for word order across concepts, applied one level down.

1.3 Lane-to-Lane Gauge Tunnels A sparse set of lane↔lane tunnels is created on co-occurrence, each carrying a CPSB-style gauge offset $\Psi_{bb'}$ and consolidated by the CPME-II phase-consistent rule:

$$\Delta w_{bb'} \;\propto\; v_b \cdot v_{b'} \cdot \cos\!\big(\theta_b - \theta_{b'} - \Psi_{bb'}\big)$$

where $\theta_b = \text{atan2}(v_b, \omega_0 x_b)$. Unlike the concept layer, $\Psi_{bb'}$ is **not** assigned from a relation table — it is *learned* as the running circular mean of observed phase differences:

$$\Psi_{bb'} \leftarrow \arg\Big(\lambda\, e^{i\Psi_{bb'}} + (1-\lambda)\, e^{i(\theta_b - \theta_{b'})}\Big)$$

This is the substrate's representation of *sequence*: the tunnel between `h` and `i` stores the typical phase lead of `h` over `i`. Nothing about letters, words, or grammar is encoded declaratively; only phase offsets between oscillators.

1.4 Byte-Level Phase-Collapse Projection Output is generated by the CPME §1.6 sweep applied to the lane population. Carrier phase ϕ_{carrier} advances monotonically across $[0, 2\pi)$ in K_{byte} steps; at each step the emitted byte is the arg-max of

$$w_b(\phi_{\text{carrier}}) \;=\; |x_b| \cdot \text{ReLU}(P_b) \cdot \exp\!\big(-\kappa_b(\phi_{\text{carrier}} - \theta_b)^2\big)$$

with P_b the lane's envelope momentum as in CPME §1.2.

Critical divergence from the concept sweep. In the concept layer, a winner is suppressed from later sweep steps — a concept appears once in a clause. In the byte layer, **winners are not suppressed**, and the sweep may make multiple passes. Letters repeat within words (`banana`, `letter`, `success`); a suppression rule would make repeated graphemes unreachable. The difference is physically motivated by what each layer represents, not introduced as an exception.

Word boundaries emerge from amplitude gaps: when the swept weight falls below the emission floor for a contiguous span of carrier phase, a boundary is emitted. Boundaries are therefore not delimiters in a grammar but silences in a resonance field.

---

2. Why This Is Not Tokens, ML, DL, or an LLM

Not tokenisation. There is no vocabulary, no merge table, no subword inventory, no `<unk>`. Input is 256 byte lanes; output is a swept byte sequence. The only inventory is the byte alphabet itself, which is a property of the encoding, not a learned artefact.

Not machine learning in the gradient sense. There is no loss function, no gradient, no backpropagation, no optimiser, no train/test split. All adaptation is the local Hebbian rule of §1.3, which uses only quantities available at the two endpoints of a single tunnel at a single tick. Nothing computes a derivative of a global objective.

Not deep learning. There is no layer stack in the DL sense — no weight matrices applied in sequence, no nonlinearity between layers, no depth over which credit is assigned. The byte layer and concept layer are two populations of coupled oscillators exchanging forces bidirectionally through one shared coupling matrix.

Not a language model. No distribution over next symbols is ever formed. There is no softmax, no sampling temperature, no autoregressive conditioning on prior output. The emitted sequence is the projection of a static field state onto a swept phase axis — a geometric readout, not a sequential prediction. Given the same field state the same sequence is produced deterministically, and the sequence is generated *simultaneously* in the sense that all lanes are already in their phases before the sweep begins.

What it is: a bidirectional associative resonance memory between a 256-oscillator sub-lexical layer and an N-oscillator conceptual layer, read out by phase-swept projection.

---

3. Where the Storage Lives (Translation Capacity Lemma)

The Translation Capacity Lemma (SI §3.1) states that any system discriminating N outcomes must store $\log_2 N$ bits somewhere. MRS does not evade this and does not claim to. The storage relocates:

Before MRS	After MRS
N label strings in `graph.nodes[].label`	$256 \times N$ byte→concept couplings $w[b][c]$
—	$\leq 256^2$ lane↔lane couplings and offsets $\Psi_{bb'}$

The relocation is the substantive change. A label table stores each surface form *atomically and independently*: `walk` and `walked` share no representation, so knowing one contributes nothing to producing the other. A coupling matrix stores forms *distributedly and compositionally*: the lanes and phase offsets that produce `walk` are largely the same ones that produce `walked`, differing by the additional offsets contributed by the co-active inflectional concept.

Generalisation to unseen forms therefore becomes possible in principle — not because information was created, but because the information was stored in a factored representation instead of an atomic one. The bit count is not reduced; its *organisation* is what changes.

---

4. Expected Dynamics

Prompt: `metformin lowers glucose`

1. **Byte injection.** Lanes for each character are driven in arrival order; t_{arrival} records the chirality gradient. Lane↔lane tunnels consolidate the observed offsets (`m` leads `e` leads `t`…).

2. **Concept resonance.** Forward couplings $w[b][c]$ excite the `metformin`, `lowers`, `glucose` cavities; propagation through the ontology raises downstream cavities (`ampk`, `glucose_uptake`).
3. **Acoustic-mass muting.** Directly-driven concepts are muted (CPME §1.3); downstream concepts are not.
4. **Reverse drive.** The surviving downstream concepts drive byte lanes through $w[b][c]^\top$. Lanes corresponding to the letters of `ampk`, `glucose`, `uptake` resonate — *without those strings being retrieved.*
5. **Byte sweep.** The carrier phase sweeps the lane population. Lanes emit in phase order, with boundaries at amplitude gaps. Output is a byte sequence assembled from resonance.

Morphological composition. If the inflectional concept for past-tense has been bound to the lanes `e`,`d` through prior exposure, and it is co-active with `lower`, the reverse drive raises both the stem lanes and the suffix lanes. The sweep — ordered by learned phase offsets — emits `lowered`, a form that need never have been stored as a label.

4b. Empirical Result: Single-Byte Lanes Are Insufficient (measured)

The single-byte formulation of §1.1–§1.4 was implemented and tested. **It does not work, and the reason is representational rather than a matter of tuning.**

Recognition was attempted by grounding every concept label to the 256 lanes at load time (binding weight = inverse byte document-frequency across the lexicon) and then recognising a query by driving its bytes and reading which concepts accumulated energy. Measured on a 2,240 -concept graph:

query	recognised as	correct
`metfromin`	`from`, `front`, `rfe`	×
`metformin`	`from`, `front`, `rfe`	×
`insluin`	`uni`, `ull`, `sil`	×
`fissoin`	`fission`	✓

Two distinct causes, and separating them matters:

(a) A normalisation error. Binding weight was divided by $\sqrt{|\text{label}|}$. For the query `metformin`, the substring `from` scored $S/2$ while `metformin` itself scored $S/3$ — the extra bytes $\{e, t, i, n\}$ are common and contribute ≈ 0 weight, but the length penalty applies regardless. A substring therefore outscored the whole word, which is why recognition failed even on *correctly spelled* input. This was a genuine bug.

(b) A representational limit. Removing the length division and replacing it with symmetric coverage (harmonic mean of query-recall and label-precision over byte sets) fixed the substring case and broke a different one: long labels sharing many bytes with the query (`intent_from_marker_label`) began to win, because a concept bound to more lanes accumulates more raw energy regardless of coverage. Each subsequent scoring adjustment relocated the failure rather than removing it.

That pattern — a defect that moves under every reweighting but never disappears — is the signature of insufficient representation. The decisive case is `metformin` versus `metfromin`: **they have identical byte sets.** A bag of distinct bytes cannot distinguish a word from its own transposition, so no weighting function over that bag can succeed. This is §5.3 (homograph collision) manifesting not as an edge case but as the dominant failure.

4b.1 Required correction: bigram lanes The sensory layer must carry ordering information. Replacing the 256 single-byte lanes with 256^2 ordered-pair lanes supplies it:

$$\texttt{metformin} \to \{\texttt{me}, \texttt{et}, \texttt{tf}, \texttt{fo}, \texttt{or}, \texttt{rm}, \texttt{mi}, \texttt{in}\}$$

$$\texttt{metfromin} \to \{\texttt{me}, \texttt{et}, \texttt{tf}, \texttt{fr}, \texttt{ro}, \texttt{om}, \texttt{mi}, \texttt{in}\}$$

Five of eight pairs are shared; three differ. The transposition is now *visible to the substrate*, and graded recognition becomes possible: a near-miss resonates strongly but not identically, which is the behaviour §1.4 requires and the single-byte layer cannot deliver.

This changes nothing about the architecture's commitments. A bigram lane is one oscillator per ordered character pair, with displacement, velocity and natural frequency — the same `ByteLane` structure, a wider array. Specifically it introduces:

- no loss function, no gradient, no fitted parameter;
- no attention, no softmax, no learned projection;
- no probability semantics — lane energy is displacement in a field.

The distinction that governs this architecture is not *"are there numbers"* but *"where do the numbers come from."* A binding weight here is **computed** from a counted structural property of the lexicon by a stated formula; changing the lexicon changes it deterministically. A weight in a trained network is **fitted** by optimising a loss and admits no explanation beyond that optimisation. Descriptive measurement is not fitted inference — measuring a mass does not make mechanics machine learning.

n-gram indexing is also not novel-to-ML prior art: it originates in 1970s information retrieval (approximate string matching, spell correction) and predates statistical learning entirely.

4b.2 Measured result after implementing bigram lanes Bigram lanes were implemented (`PhysicsEngine::bigram_bind_w`, `bind_bigram`). Grounding is identical in form to the single-byte version — inverse lane document-frequency across the lexicon — but over ordered pairs bracketed by BOS/EOS sentinels, so word-initial and word-final position are themselves observable pairs.

Recognition divides energy received by a concept by the geometric mean of its own binding mass and the query's:

$$\text{resonance}(c) \;=\; \frac{\sum_{g \in Q \cap B(c)} w_g}{\sqrt{M_c \cdot M_Q}}$$

This is energy per unit coupling — the same reason acceleration is force divided by mass — and it is why neither long nor short labels win by size. The earlier failures came from omitting exactly this normalisation.

Measured on the same 2,240-concept graph:

query	perturbation	recognised as	score	correct
`metformin`	none	`metformin`	1.000	✓
`metfromin`	transposition	`metformin`	0.686	✓
`insluin`	transposition	`insulin`	0.471	✓
`fissoin`	transposition	`fission`	0.621	✓
`fisssion`	insertion	`fission`	1.000	✓
`cavty`	deletion	`cavity`	0.830	✓
`tunel`	deletion	`tunnel`	0.885	✓

7 / 7, against 1 / 4 for single-byte lanes.

The `metformin` / `metfromin` pair is the decisive one: those two strings have **identical byte sets**, so the single-byte formulation could not distinguish them under any weighting. Over bigrams they share 5 of 8 pairs and differ in 3, and the correct concept is recovered at 0.686 against 1.000 for the exact spelling.

That gradation is the property §1.4 requires and a hash lookup cannot have: recognition degrades smoothly with distortion rather than failing discontinuously. It is also the evidence that recognition is not string matching — no comparison between the query and any stored label occurs at any point.

Status: bigram recognition is ENABLED in `codeq`, with an acceptance floor of 0.35 so unknown tokens resonate with nothing rather than matching noise. Single-byte recognition is retained in the source only as the documented negative result. Per-query telemetry reports `lexical` versus `resonant` recognition counts, so residual dependence on exact lookup is a measured quantity rather than an assumption.

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5. Failure Modes

1. **Babble before consolidation.** With near-zero couplings the lane phases are unstructured and the sweep emits non-words. This is the correct initial state of a system that has not yet learned orthography, but it will read as garbage output until the lane↔lane offsets consolidate. Consolidation requires substantial exposure — Hebbian rather than gradient, but still genuinely a training process measured in thousands of turns, not tens.
2. **Phase resolution limits word length.** With K_{byte} sweep steps the angular resolution is $2\pi/K_{\text{byte}}$. A ten-character word requires ten sufficiently separated lane phases. Long words will be truncated or slurred until K_{byte} is raised, at proportional sweep cost.
3. **Homograph collision.** A single lane per byte value means the letter `a` has one phase at any instant. In a word requiring `a` at two different positions, the multi-pass sweep of §1.4 recovers the repetition only if the two occurrences are separated by other letters whose phases bracket them. Words like `aa` or vowel-adjacent repeats are expected to fail. A positional lane bank ($256 \times K$) would resolve this at a large state cost and would reintroduce something functionally close to positional encoding; it is deliberately not proposed here.
4. **No orthographic well-formedness guarantee.** Nothing in the mechanism prevents emission of byte sequences that are not words. Well-formedness is a statistical consequence of consolidation, not a constraint. This is a genuine difference from a grammar-based generator and should be reported as such rather than patched with a dictionary filter — a filter would reintroduce the vocabulary that MRS exists to remove.
5. **Reverse-gain instability.** Bidirectional coupling through a shared matrix creates a feedback loop: concepts drive lanes drive concepts. For η_{rev} above a critical value the loop has gain greater than unity and the field diverges. The VGHT §1.3 avalanche viscosity bounds this, but η_{rev} must be tuned conservatively and the loop gain monitored.

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6. Relation to the Existing Stack

MRS is the mirror of CPME one layer down. Every mechanism it uses already exists in the substrate for the concept layer:

Mechanism	Concept layer	Byte layer (MRS)
Chirality on input	`Cavity::t_arrival`	`ByteLane::t_arrival`
Momentum filter	`env_ema` on cavities	`env_ema` on lanes
Phase definition	atan2$(v_c, \omega_0 x_c)$	atan2$(v_b, \omega_0 x_b)$
Gauge offset	Ψ from `psi_for(rel)`	Ψ *learned* from observed phase
Phase-consistent Hebbian	CPME-II on tunnels	CPME-II on lane tunnels
Phase-collapse sweep	emits concept labels	emits **bytes**
Winner suppression	yes (concept appears once)	**no** (letters repeat)

The single genuinely new element is §1.1 reverse drive. Everything else is the same physics applied to a different oscillator population — which is the strongest argument that the architecture is coherent rather than assembled.

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7. Summary

The CMD substrate was token-free on input and table-driven on output. MRS removes the remaining table by making the byte-lane layer bidirectional and equipping it with the chirality, consolidation, and phase-collapse mechanisms already built for concepts. Word forms become the projection of a byte-phase field rather than entries retrieved by graph key.

No tokeniser, no gradient, no layer stack, no next-symbol distribution. Storage obeys the Translation Capacity Lemma but is relocated from atomic surface strings into a factored coupling matrix, which is what makes morphological composition and novel forms reachable at all.

The honest cost is that consolidation is slow and early output is babble. A system that constructs its own orthography must first fail to.